

Whole sesame seed is as rich a source of mammalian lignan precursors as whole flaxseed.

The mammalian lignans enterolactone and enterodiol, which are produced by the microflora in the colon of humans and animals from precursors in foods, have been suggested to have potential anticancer effects. This study determined the production of mammalian lignans from precursors in food bars containing 25 g unground whole flaxseed (FB), sesame seed (SB), or their combination (FSB; 12.5 g each). In a randomized crossover study, healthy postmenopausal women supplemented their diets with the bars for 4 wk each separated by 4-wk washout periods, and urinary mammalian lignan excretion was measured at baseline and after 4 wk as a marker of mammalian lignan production. Results showed an increase with all treatments (65.1-81.0 $\mu\text{mol/day}$; $P < 0.0001$), which did not differ among treatments. Lignan excretion with the whole flaxseed was similar to results of other studies using ground flaxseed. An unidentified lignan metabolite was detected after consumption of SB and FSB but not of FB. Thus, we demonstrated for the first time that 1) precursors from unground whole flaxseed and sesame seed are converted by the bacterial flora in the colon to mammalian lignans and 2) sesame seed, alone and in combination with flaxseed, produces mammalian lignans equivalent to those obtained from flaxseed alone.